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SIMATS ENGINEERING

ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 10%

1. An image appears too bright. Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.
2. A noisy image contains random intensity variations. Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.
3. An image appears blurred after smoothing. Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.
4. Fine details in an image are not clearly visible. Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.
5. A noisy image must be preprocessed before further analysis. Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.
6. An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.
7. You need to separate foreground and background based on intensity. Describe and apply an appropriate segmentation method, providing a clear explanation of its process.
8. An image requires identification of object edges. Explain and justify a suitable edge detection method, describing its working and effectiveness.
9. Detected edges are noisy and fragmented. Describe and justify a method to refine edges and improve boundary detection, explaining its role in segmentation.
10. An application requires precise object boundary extraction for shape analysis. Explain and justify a suitable segmentation and boundary detection approach, describing its working and effectiveness.

90/100

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1) An image appears to bright. Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process and expected outcome.

When an image appears too bright, the negative is the most suitable gray level transformation. A negative transformation uses the gray levels of an image. Bright pixels become dark, and dark pixels become bright. This enhances details are difficult to see in overly bright image.

$$\boxed{F = S > L^{-1-r}}$$

For 8 bit $S = 255^{-r}$.

Application : Expected output.

→ Read input image. → Reduces excessive brightness

→ Replace the original pixel with mean value. → Improves contrast.

2) A noisy image contains random intensity variations explains and satisfy most suitable smoothing filter including its working mechanism & impact on noise reduction.

Concept : A medium filter is a non-linear smoothing filter that replaces that value of the center pixel with median of neighbour pixel value.

working: Select A mechanism (usually 3x3 or 5x5) arrange all pixel values in neighbourhood in ascending order. or Find the mean. replace the center pixel with mean value. Repeat the process.

Eg: (3x3) window

Original window : $\begin{bmatrix} 10 & 12 & 11 \\ 13 & 255 & 12 \\ 11 & 12 & 13 \end{bmatrix}$

Sorted values : 10, 11, 11, ~~12~~, 12, 12, 13, 13, 255.

After filtering:

$\begin{bmatrix} 10 & 12 & 11 \\ 13 & 12 & 12 \\ 11 & 12 & 13 \end{bmatrix}$ median = 12.

3) An image appears blurred after smoothing. Explain & Justify a sharpening method including working & result.

Concept: Unsharp masking sharpens a image by extracting edge info. & add it back to the original image. The mask is

mask = original − blurred.

The sharpen img is

Sharp = original + k (image) where K - sharpening factor.

working: Blur the original img from using a low-cost filter sub the blurred img from original. Obtain the edge mask. Add the mask back to the original img. Adjust the K for design enhancement

Expected result:

→ Sharper edges ⇒ better object boundary

→ Enhanced fine details → Improved img clarity

→ Increase visual quality.

4) Fine details in an img are not clearly visible. Describe & justify filtering technique used to enhance details & effects on high-frequency components

Concept: A high pass filter (hpf) is an img enhancement filter that passes high frequency comp. & suppresses low frequency comp. . This makes sharper highlights find details

Working: A high pass filter mask (kernal) is applied to the img. The filter emphasis pixels with large intensity difference from neighbours. Smooth (low-frequency) areas are reduced while edges and fine details are enhanced. The output img becomes sharper & clearer.

Eg: 3x3 (HPF) mask.

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Effect on high frequency

Enhance edges & boundary.

Highlight fine details & texture.

Increases Img. sharpness.

Expected output:

→ Sharper img.

→ Improved visibility

5) A noisy img must be preprocessed before futher analysis.

A median filter is a non-linear spatial filtering technique used to remove noise from img.

Working: Select a neighbourhood Eg: (3×3) window around the pixel. Arrange the intensity within the window in ascending order. Find the median value. Replace the center pixel with the median. Repeat the process.

Effectiveness: Removes salt & pepper noise. preserve edges & fine details improves img quality for futher processing.

6) An object is clearly brighter than the background. Explain & justify segmentation technique used.

Suitable technique: Threshold segmentation.

Threshold segmentation separates the object from the background using a single threshold intensity value

Working: Select an appropriate threshold value.

Compare every pixel with T.

If pixel intensity greater than T assign to object.

Justification:

Since the object is much brighter than the background a single threshold can accurately seperate two regions.

7) You need to seperate fore ground & background based on intensity.

Suitable method: Thresholding.

Thresholding is an intensity based segmentation Techinque used to divide an img into foreground & background based on pixel intensity.

Process:
→ choose a threshold value (T)
→ Compare each pixel intensity with T.
→ Pixels greater than T become foreground.
→ Generate binary img.

8) An img require identification of object edges. explain justify working.

Suitable method :- canny edge detection.

The canny edge detection identify object boundaries by detecting rapid intensity changes. while minimizing effects of noise.

working:

Smooth the img using gaussian filter.

Compute img gradient.

Perform non-maximum suppression to thin edge.

Apply double threshold.

Effectiveness:

Detects weak & strong edges.

Reduces thin & continuous edges. Reduce noise related noise detection.

9) Detected edges are noisy & agmented. Describe & justify.

Suitable method: Morphological operation.

Morpholigical closing is used to connect broken edge remove small gaps & produce continuous object boundary.

working:

Apply dilation to connect near by edge fragments. Apply erosion to restore the original object size.

b) An Application requires precise object boundary extraction for shape analyzes.

Suitable method: canny edge detection with threshold based segmentation.

For precised boundary extraction thresholding first separates the object from the background and canny edge detection accurately identify object boundary.

working:

→ Apply thresholding to segment the object.

→ Smooth the segmented img using Gaussian filter

→ compute intensity gradients.

→ Apply non - max suppression.

Effectiveness:

→ Accurate object segment.

→ Precise and Continuous boundaries.

→ Good localization of edges.